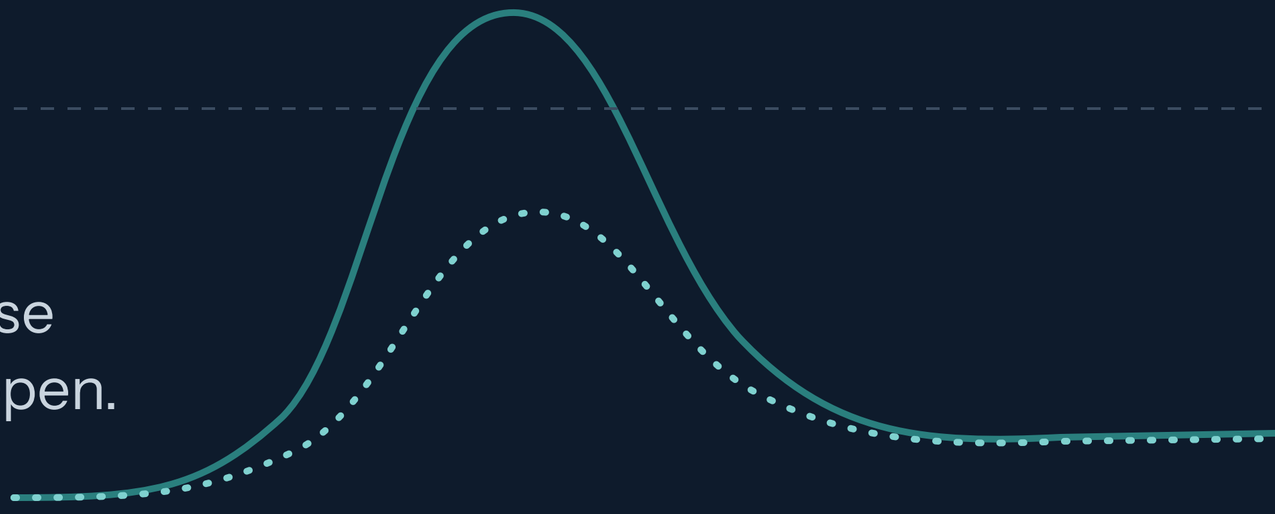
HAPPIEST HEALTH DIGITAL TWIN CHALLENGE · PHASE 1

GlucoTwin

A personal digital twin that warns of glucose spikes and lows two hours before they happen.



Team YETTO · Tashvi Wadhwa · VIT Vellore

THE PROBLEM

Diabetes in India is huge, and mostly unseen between visits

101 million

adults in India live with diabetes

ICMR-INDIAB national survey

136 million

more have prediabetes

Same survey, 2023

2–3 months

of glucose averaged into one HbA1c number

Spikes and lows in between stay invisible

Source: Anjana et al., ICMR-INDIAB, Lancet Diabetes & Endocrinology, 2023

THE GAP

Each data source sees only part of the patient

Source	What it knows	What it misses
Health record	Labs, diagnoses, medicines, history	Anything between visits
CGM sensor	Every glucose swing, every 5 min	Costs about ₹5,000 per 14 days
Smartwatch	Heart rate, HRV, steps, sleep	Glucose itself
Diary app	Meals, medicine doses, walks	How this body responds

No single source is enough. Fused, they can describe the whole patient.

CGM price: one FreeStyle Libre sensor listed at ₹5,249 by an Indian pharmacy

OUR IDEA

Fuse them into a personal digital twin

A model of one patient’s body, fitted to their own data, that can be run forward in time.

Predict

a spike or a low, two hours ahead

Explain

the reasons behind every alert

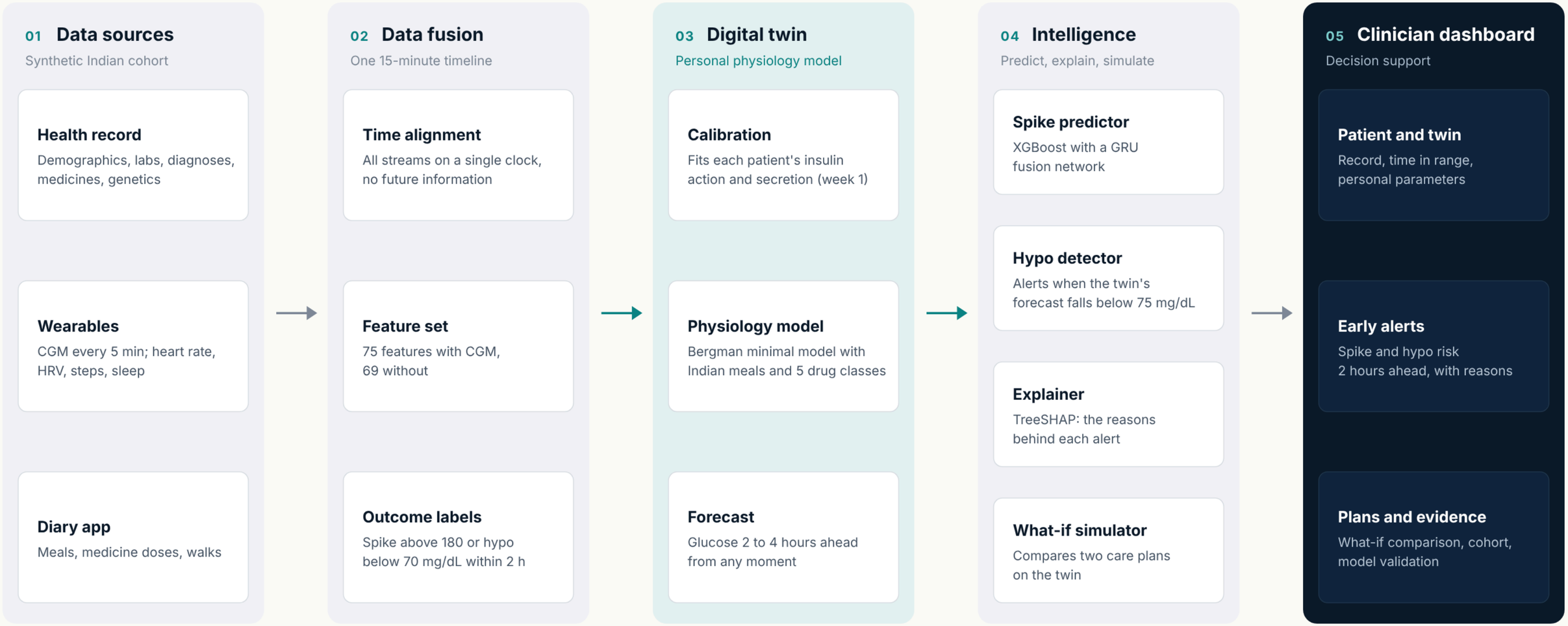
Simulate

two care plans before the patient tries either

GlucoTwin System architecture

Fusing the health record with wearable data in a personal digital twin to warn of glucose spikes and hypoglycaemia two hours ahead.

Team YETTO · VIT Vellore
Happiest Health Digital Twin Challenge, Phase 1



DATA
100% synthetic: 200 patients, 14 days each (DPDP Act 2023, HIPAA)

VALIDATION
Twin calibrated on week 1, tested on week 2 with CGM hidden; models scored on unseen patients

STACK
Python, XGBoost, PyTorch, Streamlit

LAYER 1 · DATA

A synthetic Indian cohort, generated from physiology

200
virtual patients with Type 2 diabetes

14 days
of wearable and diary data each

5 min
between CGM and smartwatch readings

36
Indian dishes, from idli to millet meals

Made for India

- Asian BMI cut-offs: overweight from 23, obese from 25
- Rice- and wheat-based regional diets
- Five common medicines, metformin to insulin glargine
- TCF7L2 gene variants and a polygenic risk score
- Missed doses driven by self-reported adherence

100% synthetic: no real patient data (DPDP Act 2023, HIPAA)

LAYER 2 · DATA FUSION

Every 15 minutes, one snapshot of the patient

01

Align

All streams on one 15-minute clock. Diary events snap to the next reading, never earlier.

02

Describe

75 features with a CGM, 69 without: glucose trend, heart rate, HRV, steps, sleep, carbs, doses, labs.

03

Label

Spike: above 180 mg/dL for 15+ minutes. Hypo: below 70 mg/dL. Both within the next 2 hours.

Every model is tested on patients it never saw in training.

LAYER 3 · THE DIGITAL TWIN

A physics model of one patient, fitted in a week

One body, as equations

An extended Bergman minimal model tracks glucose, insulin action and food in the gut, minute by minute.

- Exercise, sleep and the dawn effect
- Five medicine classes, each with its own timing
- Full state saved every 15 minutes

Week 1 · CGM worn

Calibrate four personal numbers

Insulin sensitivity, insulin secretion, liver glucose drive and drug response, found by the Cross-Entropy Method from a health-record starting guess.

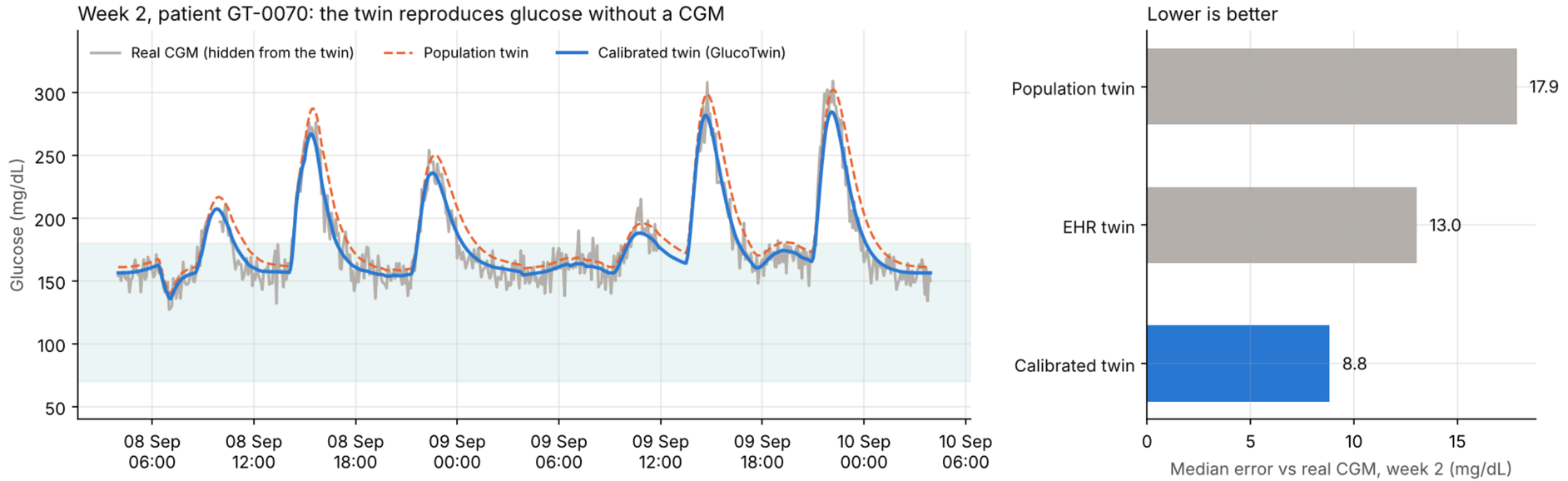
Week 2 · CGM removed

Run on cheap data

The twin follows glucose from the smartwatch, diary and health record alone.

RESULT 1 · THE TWIN

Without a CGM, the twin stays within 8.8 mg/dL



Median error in week 2 with the CGM hidden: **8.8 mg/dL**, against 13.0 for a twin set from the record and 17.9 for a one-size-fits-all twin. The key insulin measure is recovered with rank correlation **0.93**.

200 patients · disposition index = insulin sensitivity × secretion

RESULT 2 · PREDICTION

Smartwatch + twin matches a CGM on spikes

Inputs (week 2, CGM hidden unless stated)	Spike AUPRC	Hypo AUPRC
Twin alone, no machine learning	0.64	0.63
Smartwatch + health record	0.82	0.13
Smartwatch + health record + twin (GlucoTwin)	**0.86**	**0.26**
CGM worn + health record (reference)	0.84	0.19

AUPRC: 1.0 is perfect. Guessing scores 0.27 for spikes and about 0.001 for hypos, which are rare.

For lows, the twin alone is best, so GlucoTwin raises hypo alerts straight from the twin's forecast.

XGBoost, patient-level cross-validation · 135 hypo events in week 2

RESULT 3 · LOWS

Hypo alerts come straight from the twin

Rule: alert when the twin's 2-hour forecast falls below **75 mg/dL**, a safety margin above the 70 mg/dL hypo line.

91%

of hypos were warned about in advance

28%

of alerts were followed by a real hypo

0.22

false alerts per patient per day

Also tested: a GRU fusion network trained on all tasks at once. Averaged with XGBoost it lifts hypo AUPRC to **0.53** (XGBoost alone 0.35) on 50 held-out patients with a CGM.

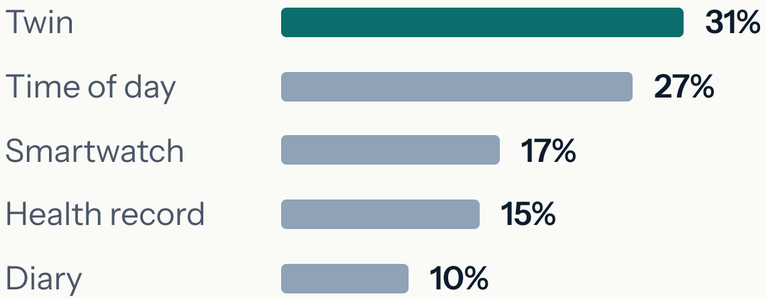
Backtest on the synthetic cohort · hypo events are rare, so these figures are uncertain

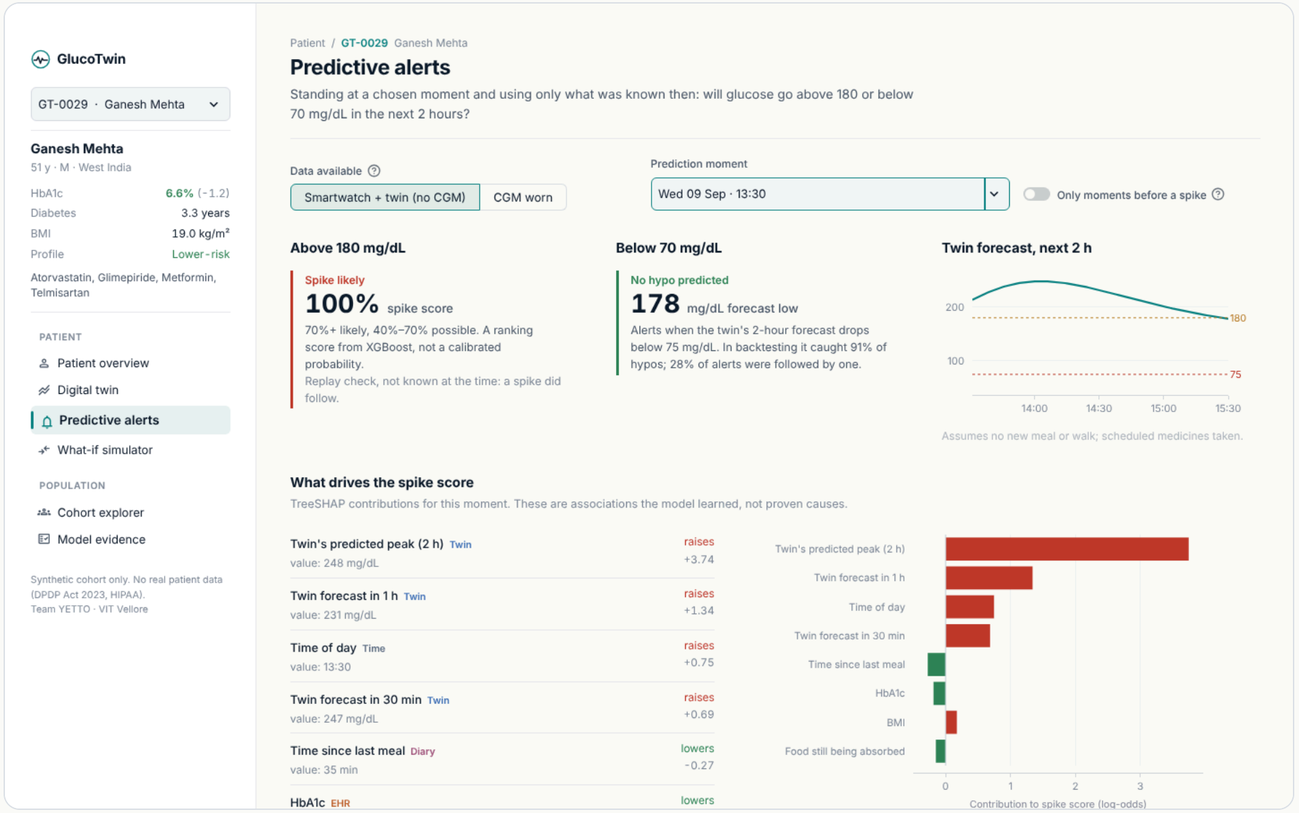
RESULT 4 · EXPLANATIONS

Every alert comes with its reasons

TreeSHAP splits each spike score into the push of every input, in plain words.

Share of the explanation, no-CGM mode

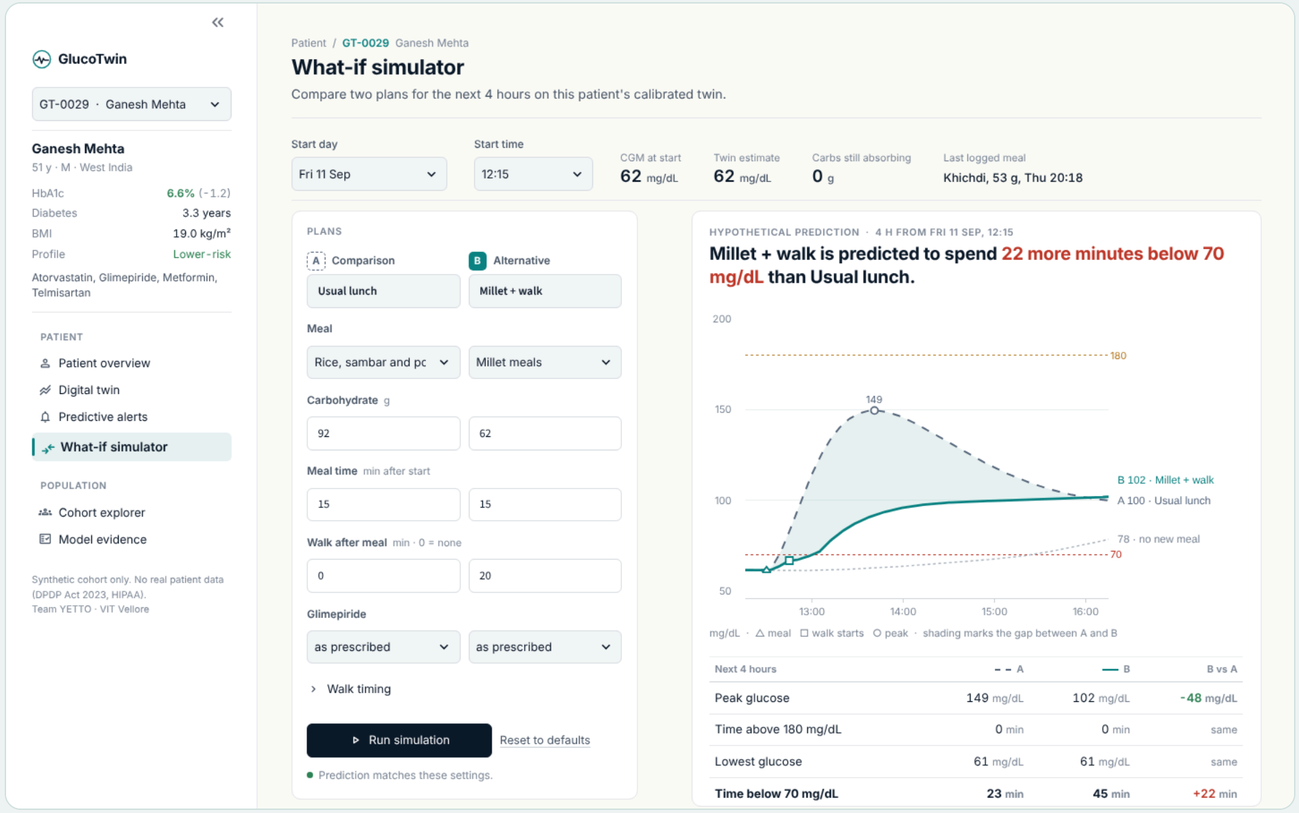




Contributions show what the model learned, not proven causes

LAYER 5 · THE DASHBOARD

Test a care plan on the twin first



1 Pick a moment

The twin starts from its saved state: glucose, insulin, food still absorbing.

2 Set two plans

Meal, carbs, timing, a walk after eating, the glimepiride dose.

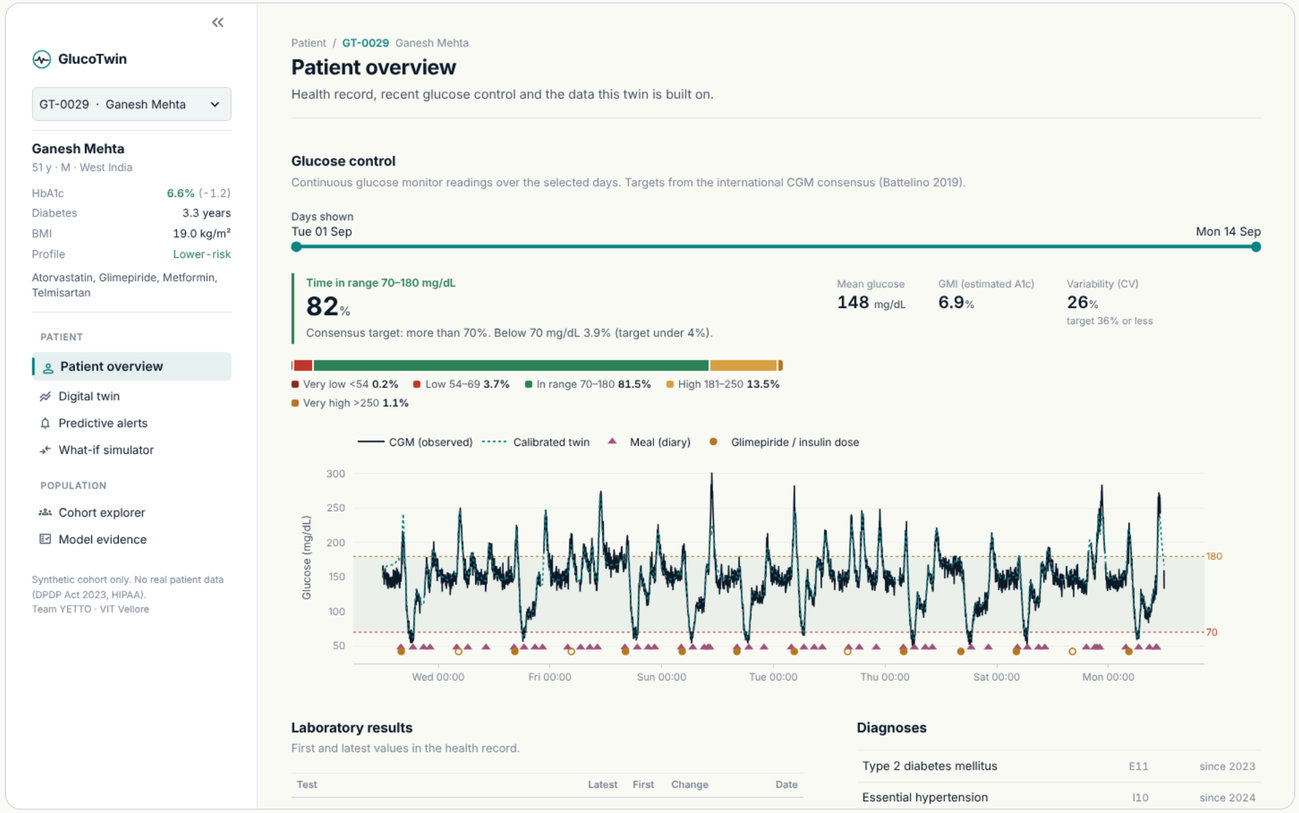
3 Compare

Peak, time above 180, lowest value, time below 70, over 4 hours.

Always labelled a hypothetical prediction, never a recommendation.

LAYER 5 · THE DASHBOARD

One dashboard, six views



Patient overview
Labs, prescriptions, time in range

Digital twin
Personal parameters, 14 days of fused data

Predictive alerts
Spike and hypo risk, with reasons

What-if simulator
Two plans compared on the twin

Cohort explorer
All 200 patients, filters, export

Model evidence
Benchmarks, figures, limitations

Every number is labelled as observed, predicted or simulated

IMPACT

One sensor to calibrate, then a smartwatch

Week 1

One CGM sensor

About ₹5,000, worn once to fit the personal twin

After that

Smartwatch + diary + record

The twin keeps forecasting spikes and lows with no sensor

Diabetes clinics

Review two weeks of alerts at each follow-up

Telemedicine

Doctors see the patient's day remotely

High hypo risk

Early warnings for people on glimepiride or insulin

Tested over one week without CGM; longer gaps need real-world validation

HONEST LIMITS

What this prototype does not prove yet

Synthetic patients only

Real wearables are noisier and people log less

Hypos are rare

135 events in week 2, so hypo figures are uncertain

The spike score ranks risk

It is not a calibrated probability

Twin and simulator share equations

This flatters the twin; real errors will be larger

Forecasts assume no new meal

An unlogged snack changes the next two hours

Decision support only

It does not diagnose, prescribe or dose

NEXT STEPS

From synthetic patients to real ones

01

Validate on real CGM and smartwatch data

Measure true errors and how often the twin needs recalibrating

02

Test the dashboard with doctors

Which alerts they act on, which they ignore, and why

03

Read records through India’s digital health stack

Standard FHIR records via the Ayushman Bharat Digital Mission

04

A companion app for patients

Easy meal logging and gentle warnings in Indian languages

GLUCOTWIN

Thank you

See the spike or the low two hours early, and test the plan on the twin first.

Team YETTO · Tashvi Wadhwa · VIT Vellore

github.com/tashviwadhwa/YETTO_VitVellore